

YERI PARK

Embedded Systems & ML Systems Engineer

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RESEARCH INTERESTS

Computer Architecture (Software–Hardware Co-Design, Compiler Design, Heterogeneous Memory Management) • ML Systems (Building deployable and efficient neural networks for real-world applications)
• Efficient AI (Quantization, Pruning, Knowledge Distillation, KV Cache Optimization)

EXPERIENCE

Hyundai Motor Company

Researcher, Department of Autonomous Driving SW Development

Seoul, Republic of Korea

Jan 2022 – Present

- Deployed INT8/FP8 mixed-precision quantization and N:M structured sparsity pruning for production autonomous driving DNNs, achieving significant reductions in real-time inference latency and memory footprint on NVIDIA Orin and Qualcomm SoCs
- Optimized VLM/VLA inference on GPU server clusters and edge devices via scalable scheduling, resource-efficient execution, and automated layer-wise quantization framework, reducing end-to-end inference overhead while maintaining task accuracy
- Architected RTOS-based autonomous driving SW frameworks targeting heterogeneous platforms (NVIDIA, Qualcomm, TI), integrating car-network interfaces (CAN/Ethernet), HILS/SILS validation environments, and sensor data loggers to ensure portability, determinism, and reliable embedded deployment
- Developed multimodal LiDAR-camera perception networks and accelerated critical kernels with CUDA and OpenCL, applying systematic profiling to identify and eliminate compute bottlenecks for real-time 3D object detection

Korea Institute of Science and Technology

Undergraduate Research Intern, Robot Vision Lab, Center for Intelligent & Interactive Robotics Jun 2021 – Sep 2021

Seoul, Republic of Korea

- Designed and implemented RGB-D CNN-based object detection and freespace perception for wearable gait-assist robots, integrated onto NVIDIA Xavier with hardware-aware acceleration for deterministic real-time performance

Hyundai Motor Company

Undergraduate Research Intern, Department of LiDAR Perception Development Sep 2021 – Dec 2021

Seoul, Republic of Korea

- Built high-performance LiDAR preprocessing pipelines, ground segmentation, and point-cloud clustering algorithms; implemented real-time PCD visualization tool for algorithm validation and debugging

Ewha Womans University

Undergraduate Research Intern, Digital Signal Processing Lab

Seoul, Republic of Korea

Jun 2019 – Aug 2019

- Developed human respiration classification model using IR-UWB radar signals with comprehensive data pre-processing and feature extraction

EDUCATION

Georgia Institute of Technology

Master of Science in Computer Science

Online

Sep 2025 – Present

Ewha Womans University

Bachelor of Science in Electronic and Electrical Engineering, GPA: 97.9/100 Mar 2017 – Feb 2022

Seoul, Republic of Korea

SKILLS

Languages: Korean, English

Programming: Modern C/C++, Python (PyTorch), Rust, CUDA/OpenCL, Haskell

ML Tools: TensorRT, ONNX-MLIR, Swin Transformer, PyTorch, OpenCV

Platforms: NVIDIA Orin, Qualcomm SoCs, NVIDIA Xavier, NVIDIA A100

Certifications: ISTQB SW Testing CTFL, ISTQB AI Testing, NVIDIA CUDA Programming

Relevant Courses: Computer Architecture, Embedded Systems, Operating Systems, Deep Learning, Integrated Circuits, GPU HW and SW

PROJECTS

Edge AI Network Pruning Project

Aug 2025

- Applied multiple (un)structured pruning techniques to computer vision networks and integrated them on NVIDIA Orin embedded platforms
- *Tools:* C/C++, NVIDIA TensorRT-LLM, Swin Transformer

Rust-based ONNX-MLIR Optimization Project

Apr 2025

- Built a network optimization framework using ONNX-MLIR and Rust for universal hardware compilation
- *Tools:* Rust, Python, C++, ONNX-MLIR

Haskell Monad-based Sensor Data Logger Project

Jul 2024

- Developed a resource-safe sensor data logger using Haskell monads to explore functional programming paradigms
- *Tools:* Haskell

Autonomous Driving LiDAR Perception Project

Dec 2023

- Implemented LiDAR perception modules: ground segmentation, curb detection, free-space estimation, and 3D object detection
- *Tools:* C/C++, CUDA, OpenCL, PyTorch, LiDAR

Wearable Robot Perception Project

Dec 2021

- Developed walkable path detection via ground semantic segmentation networks for wearable robot perception
- *Tools:* C/C++, Python/PyTorch

HONORS & AWARDS

2nd Prize, Ewha Engineering Capstone Design Contest

Dec 2020

Developed an AI-based audience response analysis system in video presentation environment leveraging Face Recognition Network, OpenCV, and WebRTC

Excellence Award, Korea Electric Power Engineering Camp

Aug 2020

Recognized as Top 3 and scholarship recipient at national engineering camp

Academic Excellence Scholarship (4 consecutive semesters)

2017 – 2021

Awarded for outstanding academic performance at Ewha Womans University

Jump Mentoring Scholarship Recipient

Dec 2017

Participated as tutor in youth mentoring program